

Symphytum officinale (Boraginaceae)

Comfrey, Knitbone

Comfrey's names testify to its traditional use in mending broken bones. "Comfrey" is a corruption of *con firma*, meaning the bone is "made firm," *Symphytum* is derived from the Greek for "to unite," and knitbone speaks for itself. Comfrey is also a wound herb. K'Eogh in his *Irish Herbal* (1735) wrote that it "heals all inward wounds and ruptures." Today, it is still highly regarded for its healing properties.



Comfrey was known to the Greek physician Dioscorides in the 1st century CE, who wrote about it in his *Materia Medica*.

Habitat & Cultivation

An indigenous European plant, comfrey grows in all temperate regions of the world, including western Asia, North America, and Australia. It thrives in moist, marshy places. It can be grown from seed in spring or by root division in autumn, and the leaves and flowering tops are harvested in summer. The root is unearthed in autumn.

Key Constituents

- Allantoin (up to 4.7%)
- Mucilage (about 29%)
- Triterpenoids
- Phenolic acids (rosmarinic acid)
- Asparagine
- Pyrrolizidine alkaloids (0.02–0.07%)
- Tannins

Key Actions

- Demulcent
- Astringent
- Anti-inflammatory
- Heals wounds and bones

Research

■ **Active constituents** Allantoin is a cell-proliferant that stimulates the repair of damaged tissue. The herb's anti-inflammatory activity is partly due to the presence of rosmarinic acid and phenolic acids.

■ Pyrrolizidine alkaloids

Research shows that, as isolated substances, certain pyrrolizidine alkaloids can be highly toxic to the liver. It is still unclear whether they are toxic in the context of the whole plant, as they are only present in minute amounts, often being completely absent from samples of dried aerial parts. The highest concentration is in the root, which should not be used internally. Skin applications, as well as the aerial parts of the plant, are considered safe.

■ **Clinical research** In Germany and elsewhere in Europe, comfrey is widely used for sprains, bruises, and sports injuries. Research, mostly in Germany, endorses the traditional knowledge of comfrey's wound-healing ability. In a 2007 study, physicians rated the efficacy of a comfrey leaf cream in healing abrasions. The doctors rated its effectiveness as good or very good in 93% of cases, and complete healing took 4 days with comfrey and 7 days with placebo. Other studies indicate comfrey's value in promoting tissue repair and as an anti-inflammatory in conditions such as sprained ankle, osteoarthritis, and lower back pain.

Traditional & Current Uses

■ **Injuries** Comfrey's ability to promote the healing of bruises,



Comfrey

A perennial growing to 3 ft (1 m), with thick leaves and bell-like white to pink or mauve flowers.

Parts Used

Root is harvested in autumn when the allantoin levels are highest.



Fresh root



Dried root



Fresh aerial parts



Dried aerial parts

Aerial parts are rich in anti-inflammatory and astringent substances.

Key Preparations & Their Uses

⚠ **Cautions** Do not use on dirty wounds as rapid healing can trap dirt or pus. Take internally only under professional supervision. Restricted in some countries. Do not take during pregnancy or while breastfeeding.



Infused oil of leaves (to make, p. 294). Apply to sprains.



Chopping leaves For boils, apply as a poultice (see p. 295).



Ointment of leaves (to make, p. 295). Apply to bruises.

🌿 **Tincture** of root (to make, p. 292). Apply undiluted to acne.

sprains, fractures, and broken bones has been known for thousands of years. It encourages ligaments and bones to knit together firmly. A comfrey compress applied immediately to a sprained ankle can significantly reduce the severity of the injury. The combination of tannins and mucilage helps to soothe bruises and grazes.

■ **Other uses** Comfrey preparations have many other uses and can be

applied to heal problems such as insect bites, scars, skin inflammation, acne, and mastitis.

Self-help Uses

- **Acne & boils**, p. 305.
- **Fractures**, p. 312.
- **Fungal skin infections**, p. 304.
- **Healing wounds**, p. 304.
- **Inflamed skin rashes**, p. 303.
- **Stiff & aching joints**, p. 313.